

Figure 2: Using dpkg to list the kernel source files

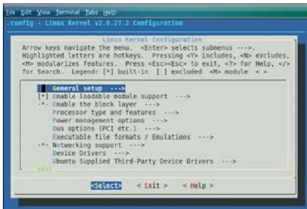


Figure 3: Configuring the kernel in Ubuntu

supported by the kernel. And if you are an intermediate-level user you might want to take away some 'extra' features (and support) from the kernel and build a custom one with just the essentials.

Well, we cannot address patching at this stage. But we can speak about how customisation works. Please note that this demonstration is on an OS derived from Debian.

In Segment 1, we discussed commands like *uname*. Now let's have:

```
aasisvinayak@GNU-BOX:~$ uname -r
2.6.27-7-generic
```

We've got the version number. Now let's get the source code by issuing the following command:

```
sudo apt-get install linux-source-2.6.27 kernel-package libncurses5-
dev fakeroot
```

You can use *dpkg* to show you the files. Please refer to the screenshot in Figure 2.

You can also see that the source is located at `/usr/src/`.

```
aasisvinayak@GNU-BOX:~$ dpkg -L linux-source-2.6.27
```

```
/usr
/usr/src
/usr/src/linux-source-2.6.27.tar.bz2
/usr/share
/usr/share/doc
/usr/share/doc/linux-source-2.6.27
/usr/share/doc/linux-source-2.6.27/copyright
/usr/share/doc/linux-source-2.6.27/changelog.Debian.gz
```

Now, as the root user (you may use: `sudo /bin/bash`), execute the following:

```
cd /usr/src
bunzip2 linux-source-2.6.27.tar.bz2
tar xvf linux-source-2.6.27.tar
ln -s linux-source-2.6.27 linux
```

Then, let's make a copy of the existing kernel configuration by issuing the following:

```
cp /boot/config-`uname -r` /usr/src/linux/.config
```

Now let's start customising it:

```
root@GNU-BOX:/usr/src# cd /usr/src/linux-headers-2.6.27-7
root@GNU-BOX:/usr/src/linux-headers-2.6.27-7#
root@GNU-BOX:/usr/src/linux-headers-2.6.27-7# make menuconfig
HOSTCC scripts/basic/fixdep
HOSTCC scripts/basic/docproc
HOSTCC scripts/kconfig/conf.o
scripts/kconfig/conf.c: In function 'conf_askvalue':
scripts/kconfig/conf.c:104: warning: ignoring return value of 'fgets',
declared with attribute warn_unused_result
scripts/kconfig/conf.c: In function 'conf_choice':
scripts/kconfig/conf.c:306: warning: ignoring return value of 'fgets',
declared with attribute warn_unused_result
HOSTCC scripts/kconfig/kxgettext.o
HOSTCC scripts/kconfig/ldialog/checklist.o
HOSTCC scripts/kconfig/ldialog/inputbox.o
HOSTCC scripts/kconfig/ldialog/menubox.o
HOSTCC scripts/kconfig/ldialog/textbox.o
HOSTCC scripts/kconfig/ldialog/util.o
HOSTCC scripts/kconfig/ldialog/yessno.o
HOSTCC scripts/kconfig/mconf.o
SHIPPED scripts/kconfig/zconf.tab.c
SHIPPED scripts/kconfig/lex.zconf.c
SHIPPED scripts/kconfig/zconf.hash.c
HOSTCC scripts/kconfig/zconf.tab.o
In file included from scripts/kconfig/zconf.tab.c:2486:
scripts/kconfig/confdata.c: In function 'conf_write':
scripts/kconfig/confdata.c:501: warning: ignoring return value of 'fwrite',
declared with attribute warn_unused_result
scripts/kconfig/confdata.c: In function 'conf_write_autoconf':
scripts/kconfig/confdata.c:739: warning: ignoring return value of 'fwrite',
declared with attribute warn_unused_result
```